

01 Activity 1

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Section	BSIT-A508	Score	

Instructions: (2 items x 15 points)

1. Provide one (1) traditional application, and one (1) advanced application of information systems. It should be different from the examples mentioned in the handout.

Traditional application: Transaction Processing System (TPS)

Example: **Point of Sale (POS) System**

A POS system records daily transactions in retail or service environments, such as sales, returns, and inventory management. It's used to process customer purchases, track product sales, and manage financial reporting.

Advance application: Artificial Intelligence (AI)-Powered Decision Support System (DSS)

Example: **Predictive Analytics in Healthcare**

AI-driven DSS uses machine learning algorithms to analyze large datasets and make informed decisions or predictions, such as predicting patient outcomes, disease risks, or treatment plans.

2. Identify each component of information systems used in your provided applications. See **Page 2 of 01 Handout 1** for examples.

Traditional Application				
Hardware	Software	Data	People	Process
• POS terminal or computer. • Barcode scanner • Receipt printer. • Cash drawer. • Card reader (for credit/debit payments)	• POS software (e.g., Square POS, Lightspeed, or proprietary POS systems). • Operating system (e.g., Windows, Linux, or macOS). • Database management system (DBMS) to store transaction records.	• Sales transactions (e.g., items purchased, prices, quantities). • Inventory data. • Customer data (if integrated with loyalty programs). • Financial data (payments, refunds).	• Cashiers and store clerks who operate the POS. • Store managers and supervisors who analyze reports. • IT staff for maintenance and updates.	• Scanning items and processing payments. • Automatically updating inventory levels after each sale. • Generating sales reports and tracking daily transactions. • Processing returns and refunds.

Advanced Application				
Hardware	Software	Data	People	Process
• High-performance servers for data processing. • Cloud computing infrastructure for storage and computation. • Workstations or computers for clinicians and decision-makers. • Medical devices and IoT sensors (for collecting real-time patient data).	• AI and machine learning platforms (e.g., TensorFlow, IBM Watson Health). • Electronic Health Record (EHR) systems. • Specialized DSS software for healthcare (e.g., clinical decision support tools). • Database management system (DBMS) for handling large datasets.	• Patient health records (e.g., medical history, diagnostics). • Real-time clinical data (e.g., vital signs, lab results). • Research data (e.g., drug effectiveness, treatment outcomes). • Predictive models generated by AI algorithms.	• Clinicians (e.g., doctors, nurses) who use the DSS to make treatment decisions. • Data scientists and AI experts who develop and train predictive models. • IT and support staff who manage system integration and performance.	• Collecting and inputting patient data into the system. • Running AI algorithms to analyze patient data and generate predictions. • Providing clinicians with actionable insights (e.g., diagnosis, treatment suggestions). • Continuous monitoring and updating of predictive models based on new data.

GRADING RUBRIC:

Performance Standards	Points
The student provided a unique application of information systems.	5
The student identified the five (5) components of information systems.	10
Total	15